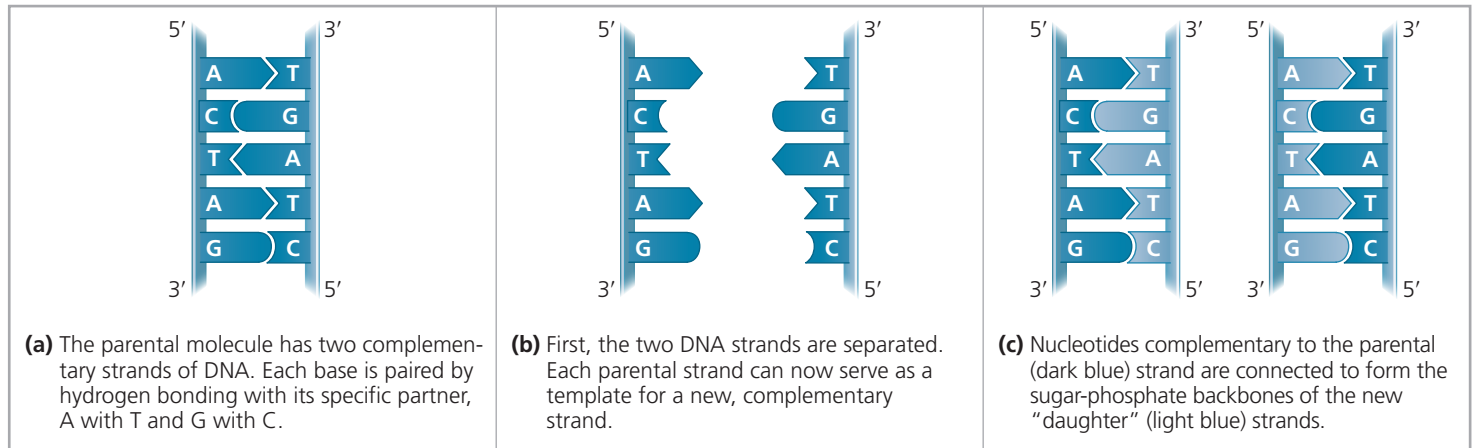


▼ **Figure 16.10 A model for DNA replication: the basic concept.** In this simplified illustration, a short segment of DNA

has been untwisted. Simple shapes symbolize the four kinds of bases. Dark blue represents DNA strands present in the parental molecule;

light blue represents newly synthesized DNA.

➔ **Mastering Biology Animation:**
DNA Replication: An Overview



The Basic Principle: Base Pairing to a Template Strand

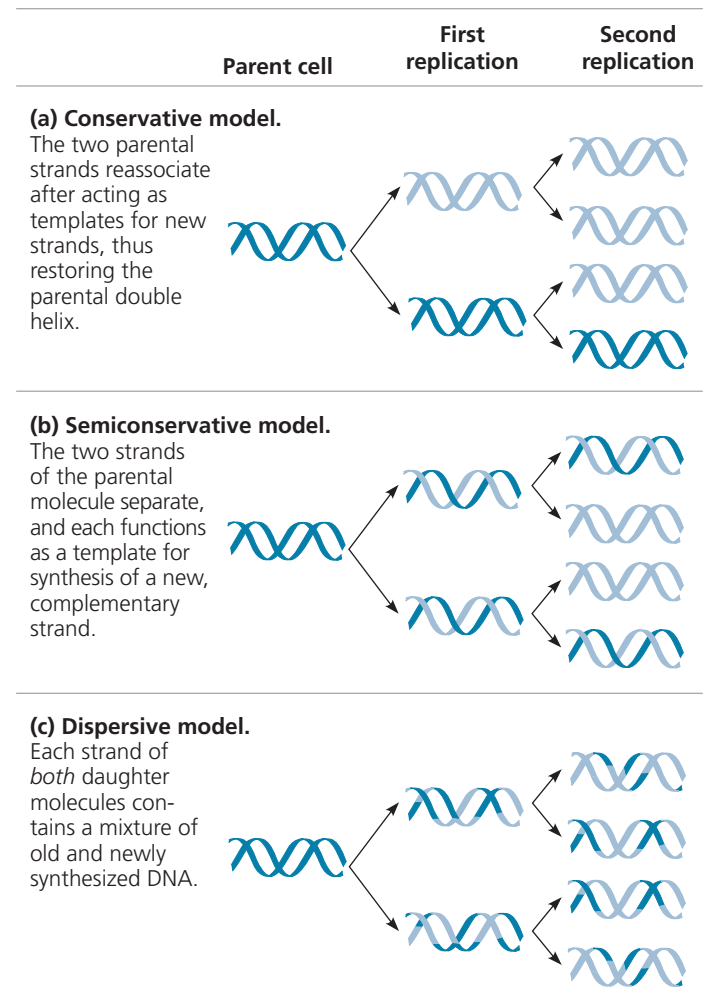
In a second paper, Watson and Crick stated their hypothesis for how DNA replicates: "Now our model for deoxyribonucleic acid is, in effect, a pair of templates, each of which is complementary to the other. We imagine that prior to duplication the hydrogen bonds are broken, and the two chains unwind and separate. Each chain then acts as a template for the formation on to itself of a new companion chain, so that eventually we shall have two pairs of chains, where we only had one before. Moreover, the sequence of the pairs of bases will have been duplicated exactly."²

Figure 16.10 illustrates the basic idea. If you cover a DNA strand in **Figure 16.10a**, its linear sequence of nucleotides is revealed by applying the base-pairing rules to the uncovered strand. The two strands are complementary; each stores the information necessary to reconstruct the other. When a cell copies a DNA molecule, each strand serves as a template for ordering nucleotides into a new, complementary strand. Nucleotides line up along the template strand according to the base-pairing rules and are linked to form the new strands. One double-stranded DNA molecule becomes two, each an exact replica of the "parental" molecule.

This model of DNA replication remained untested for several years following publication of the DNA structure. The necessary experiments were simple in concept but difficult to perform. Watson and Crick's model predicts that when a double helix replicates, each of the two daughter molecules will have one old strand, from the parental molecule, and one newly made strand. This **semiconservative model** can be distinguished from a *conservative model* of replication, in which the two parental strands somehow come back together after the process (that is, the parental molecule is conserved). In yet a third model, called the *dispersive model*, all four strands of DNA following replication have a mixture of old and new DNA (**Figure 16.11**).

▼ **Figure 16.11 DNA replication: three alternative models.**

Each short segment of double helix symbolizes the DNA within a cell. Beginning with a parent cell, we follow the DNA for two more generations of cells—two rounds of DNA replication. Parental DNA is dark blue; newly made DNA is light blue.



²J. D. Watson and F. H. C. Crick, Genetical implications of the structure of deoxyribonucleic acid, *Nature* 171:964–967 (1953).